

Vijay Stock Exchange (VSE) – Angular + Spring Boot JWT Project Summary

This document summarizes the complete frontend authentication and routing architecture implemented using Angular 21 and Spring Boot with JWT authentication. The explanation is tailored for interview preparation and real-world understanding.

1. Technology Stack

- 1 Frontend: Angular 21 with Angular Material
- 2 Backend: Spring Boot (REST APIs)
- 3 Authentication: JWT (JSON Web Token)
- 4 Authorization: Route Guards + JWT Filters

2. Application Flow

- 1 User opens application and sees Login page first
- 2 User submits credentials
- 3 Spring Boot validates credentials and generates JWT
- 4 Angular stores JWT in localStorage
- 5 User is redirected to Dashboard
- 6 All protected API calls send JWT automatically
- 7 If token is missing or expired, user is redirected to Login

3. Layout Architecture

- 1 AuthLayout: Used only for Login page (no navbar, no footer)
- 2 MainLayout: Used after login (navbar, footer, content)
- 3 This separation improves security and user experience

4. Authentication Service (Angular)

- 1 Handles login API call
- 2 Stores JWT token in localStorage
- 3 Provides isLoggedIn() method for guards
- 4 Handles logout by clearing token

5. Route Guard

- 1 Prevents unauthenticated access to protected routes
- 2 Checks token presence using AuthService

- 3 Redirects user to login if not authenticated

6. HTTP Interceptor

- 1 Automatically attaches JWT token to every HTTP request
- 2 Adds Authorization header: Bearer <token>
- 3 Centralizes token handling

7. Backend JWT Authentication

- 1 Spring Boot exposes /auth/login endpoint
- 2 Validates credentials
- 3 Generates signed JWT with expiry
- 4 Backend remains stateless

8. Why JWT?

- 1 Stateless authentication
- 2 Scales well with microservices
- 3 No server-side session storage
- 4 Industry standard security approach

9. Logout Flow

- 1 Token removed from browser storage
- 2 User redirected to login page
- 3 Protected routes blocked immediately

10. Interview One-Line Summary

I implemented a secure Angular application using JWT-based authentication with route guards, HTTP interceptors, and Spring Boot backend validation. The architecture is scalable, secure, and production-ready.